MENOFIA UNIVERSITY

Faculty of Computers and Information

Software Engineering 2, Sections 3 and 4

Weeks 3–4: Defect Reports, First Selenium Script, Locators and Assertions

Student name	Academic number

One task for weeks 3 and 4, done on your own laptop. Part A needs only a browser. Parts B and C need your own Maven project. You hand it in twice, on the form:

	Hand in	When
Parts A and B	This sheet with Parts A and B, ShopVisit.java, one screenshot of a run	Before section 5. Run Part B in front of the TA at the start of section 5.
Part C	This sheet with Part C, ShopCheck.java, one screenshot of a run	Section 5 explains CSS, XPath and assertEquals. Hand in before section 6 and run it in front of the TA at the start of section 6.

Part A: Test the shop by hand

The site	https://www.saucedemo.com
Password	secret_sauce for every user
standard_user	The shop working correctly. This is your expected behaviour.
problem_user	The same shop, with defects in it.
Chrome warning	If Chrome says the password was found in a data breach, close the warning. It is a public practice password.

Exercise 1: Hunt for defects

Section 3 · slides 4, 9 and 19

Log in as standard_user and use the shop: sort the products, open a product, add things to the cart, remove one, start a checkout. Then log out (menu ☰ › Logout), log in as problem_user and do exactly the same things. Try the menu (☰) too. Every difference is a defect.

problem_user has eight defects. The first one is written for you as an example. Find the other seven.

#	What you did	Expected (standard_user)	Actual (problem_user)
1	Example: looked at the products page	Each product has its own photo	All six products show the same photo of a dog
2			
3			
4			
5			
6			
7			
8			

Exercise 2: Write the full report

Section 3 · slides 6 to 13

Pick the defect from Exercise 1 that hurts the business most. Write the report so a developer who has never seen this site can make it happen again.

Field	Your report
Title	
Steps	1. 2. 3. 4. 5.
Expected	
Actual	
Environment	Type chrome://version in the address bar to find your Chrome version.
Severity	
Priority	
Attachment	Take a screenshot and draw a red box around the problem. File name:

Exercise 3: You decide

Section 3 · slide 11

The release is tomorrow. The developer has time to fix only two of your defects. Which two, and which one waits?

	Defect (# from Exercise 1)	Why
Fix first		
Fix second		
Leave for later		

Exercise 4: The life of your report

Section 3 · slides 14 to 16

Your report from Exercise 2 is in the system. Which state is it in now?

#	What happens	State
1	The developer says another student reported the same defect this morning.	
2	The developer fixed it. The new build is on the test server, but you have not tried it yet.	
3	You log in as problem_user on the new build and it still happens.	

Part B: Automate it

Exercise 5: Your own project

Section 3 · slides 26 and 27

Create your own Maven project in IntelliJ (File › New › Project › Java, Build system: Maven). Add the Selenium and TestNG dependencies to pom.xml and reload Maven. Then write FirstProject from the section and run it.

Your Java version	Selenium version in External Libraries	What did FirstProject print?

Exercise 6: Write ShopVisit

Section 3 · slides 27 to 31

In the same project, write a class called ShopVisit with these six methods. Use only what we used in the section.

Method	Annotation	What it does
prepare	@BeforeTest	Opens Chrome.
openShop	@Test(priority = 1)	Opens https://www.saucedemo.com/ and prints the title and the current URL.
goAndBack	@Test(priority = 2)	Goes to https://www.selenium.dev/ with navigate().to(), goes back, and prints the title.
devices	@Test(priority = 3)	Makes the window phone size (390 × 844), then tablet size (1366 × 1024), then maximizes it. Wait two seconds between the steps so you can see them.
newTab	@Test(priority = 4)	Prints the window handle, opens a new tab, opens https://www.google.com/ in it, prints the new handle.
tearDown	@AfterTest	Closes the browser.

Run it, then copy what it printed.

What was printed	Your output
Title (openShop)	
URL (openShop)	
Title after going back	
First tab handle (first 6 characters)	
Second tab handle (first 6 characters)	

Exercise 7: Change it and watch

Section 3 · slides 29 and 31

Make each change, run ShopVisit, write what you saw, then undo the change before the next one.

#	Change	What did you see, and why?
a	Swap the priorities of openShop and goAndBack (1 and 2). What does the title after going back print now?	
b	In goAndBack, replace navigate().to(...) with driver.get(...). Did going back still work? What does that tell you?	
c	Delete the tearDown method. What do you see when the run finishes?	

Part C: Find it and check it

Do this part after section 5. Hand it in before section 6.

Exercise 8: Write the locators

Sections 4 and 5 · slides 5 to 13

On saucedemo.com, open DevTools (F12) and write one locator for each element. Test it with Ctrl + F in the Elements tab and write how many matches it found.

#	Element	Your locator	Type	Matches
1	The Username field, using its name			
2	The Login button, without using its id			
3	The error message after clicking Login with both fields empty			
4	The word Products after you log in			
5	The Add to cart button of Sauce Labs Onesie			
6	The number on the cart after you add one item			

Exercise 9: Automate your own defects

Section 5 · slides 14 to 21

In the same project, write a class called ShopCheck:

Part	What it does
String user = "standard_user";	A field at the top. You will change only this line.
@BeforeTest	Opens Chrome in incognito (the three lines below) and logs in as user, using three different locator types.
Three @Test methods	Each one checks one defect you found in Exercise 1 with Assert.assertEquals(actual, expected). Use getText() or getAttribute("value") to read the actual result.
@AfterTest	Closes the browser.

Open Chrome like this. Without it, Chrome shows a "password found in a data breach" box after the login, and every click after that does nothing:

```
ChromeOptions options = new ChromeOptions();
options.addArguments("--incognito");
driver = new ChromeDriver(options);
```

Run it as standard_user: all three must pass. Then change the field to problem_user and run it again.

Test method	Defect # (Exercise 1)	Passed as standard_user?	Failure message as problem_user